

GEOGRAPHY

PHYSICAL GEOGRAPHY

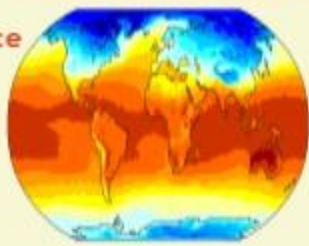


CLIMATOLOGY

MODULE – 14

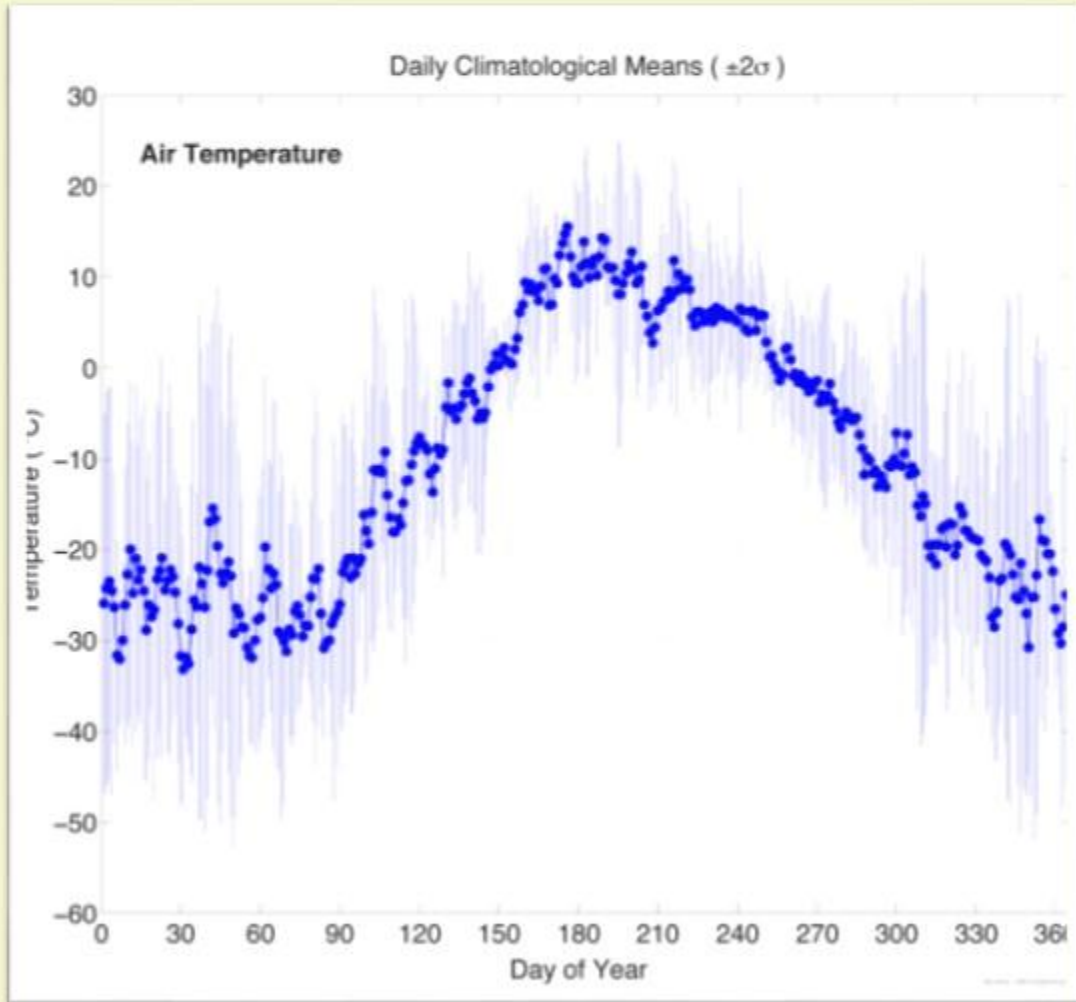
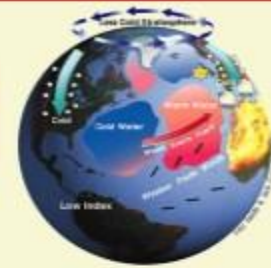
TEMPERATURE

- MEAN TEMPERATURE
- DISTRIBUTION OF TEMPERATURE

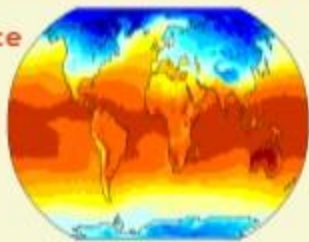


DISTRIBUTION OF TEMPERATURE

MEAN TEMPERATURE

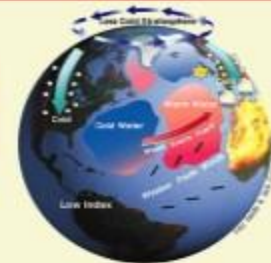


- 1 Earth receives maximum energy at 12 noon but the **maximum temperature never occurs at 12 noon** because the transformation of solar energy into heat requires time.
- 2 Max temp received at 2-4 PM.
- 3 The highest temperature recorded within 24 hours is called **maximum daily temperature**.
- 4 Lowest temperature recorded between 4-5 AM instead of midnight. This is known as the **minimum daily temperature**.
- 5 The average of maximum and minimum temperature within 24 hours is called **mean daily temperature**.



DISTRIBUTION OF TEMPERATURE

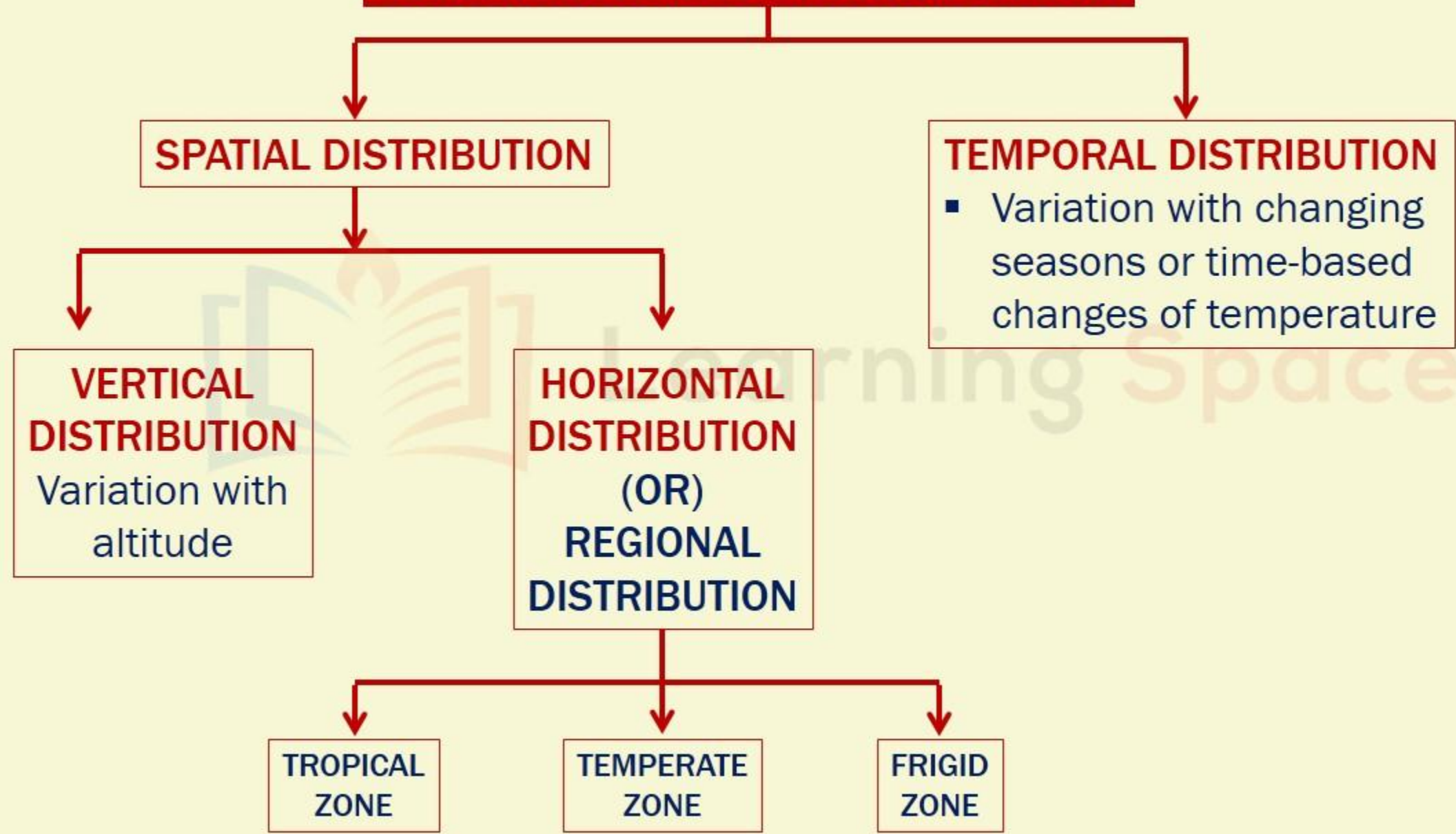
MEAN TEMPERATURE

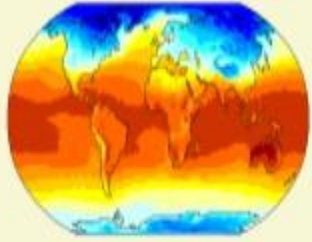


Diurnal range of temperature = (Daily maximum – Daily minimum)
(24 hrs)

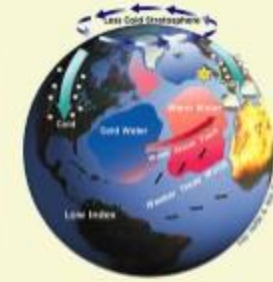
Annual range of temperature = (Annual maximum – Annual minimum)
(365 days)

DISTRIBUTION OF TEMPERATURE

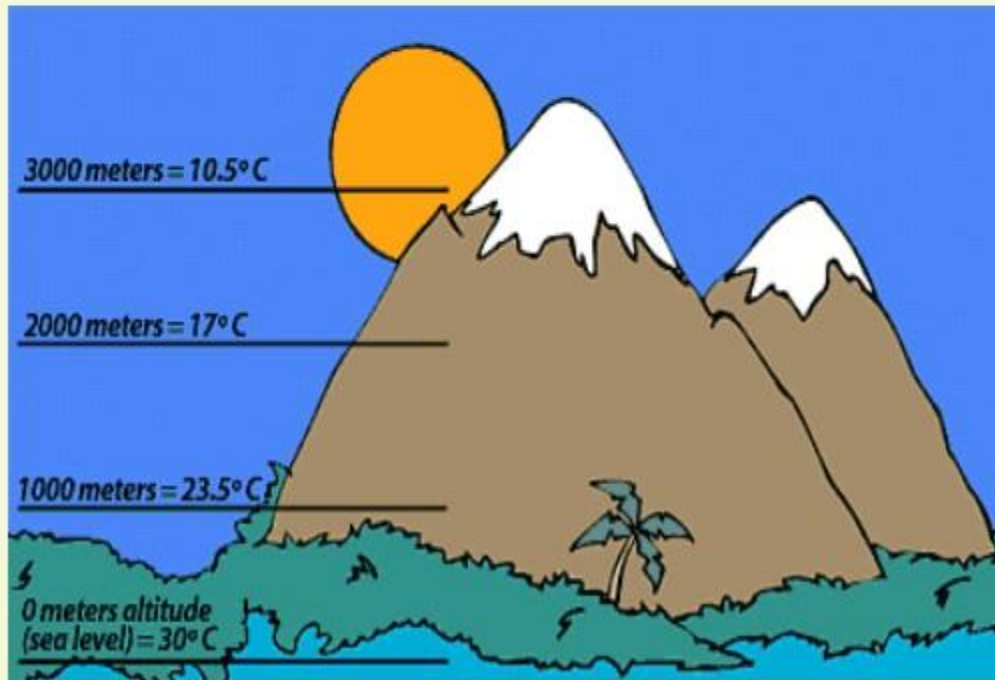




VERTICAL DISTRIBUTION OF TEMPERATURE



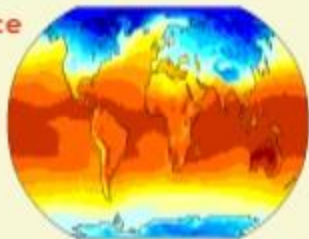
- 5** Temperature drops with increasing altitude at the rate of 6.5 degree Celsius per 1000m. This rate is known as **Normal Lapse Rate(NLR)**.



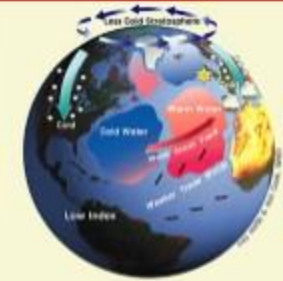
Why does this temperature drop happen?

- 1** As we ascend higher in the atmosphere, the **amount of heat gets lost to successive layer** lying below.
- 2** The air pressure is higher in the lower portion of the atmosphere near the earth's surface because of the weight of all the overlying layers above. Thus **density of air decreases** as we move upwards.
- 3** The **lower layer of the air contains more water vapour and dust particles** than the layers above and hence it **absorbs larger amount of heat radiated** from the earth's surface than the upper layers.





HORIZONTAL DISTRIBUTION OF TEMPERATURE



WHAT ARE ISOTHERMS?

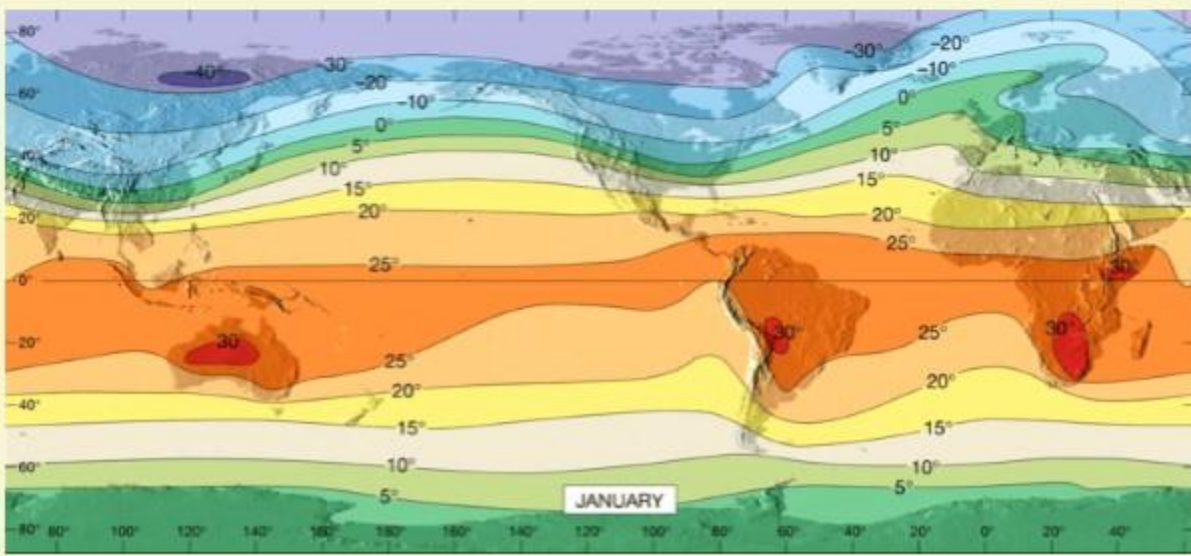
The lines drawn on maps joining the places of equal temperature reduced to sea level are called isotherms

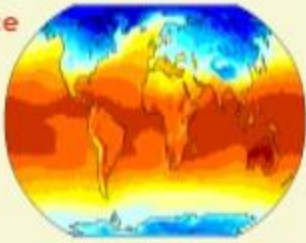
WHY ARE THEY REDUCED TO SEA LEVEL?

This is done to negate the effects of altitude when drawing isotherms of a place

DOES IT REPRESENT THE ACTUAL TEMPERATURES OF A PLACE?

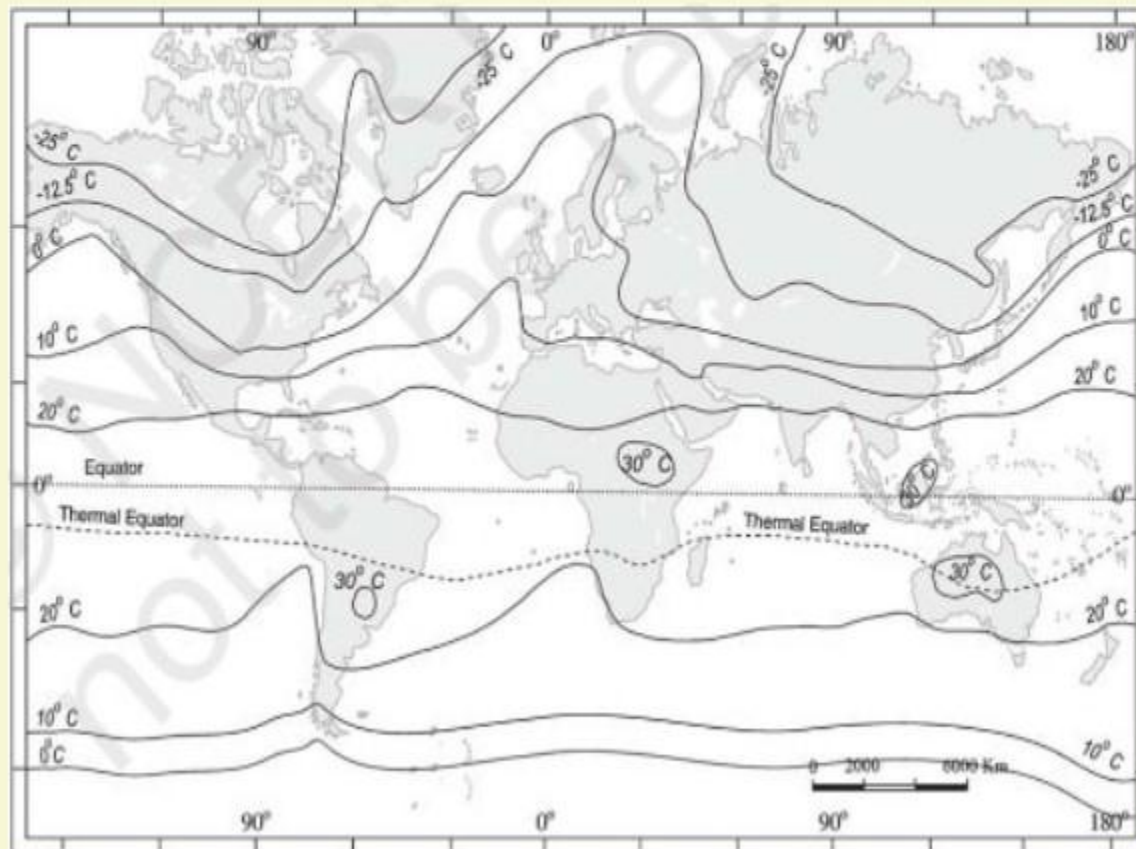
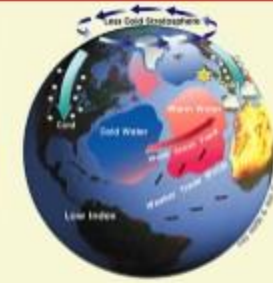
- NO! It does not show the real temperature of the places through which they pass rather they show the temperature of the places reduced to sea level.
- This is why the isotherm maps are not useful for farmers because they need real temperature of a particular place for growing crops.





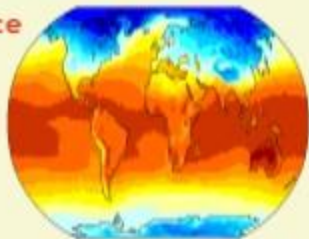
HORIZONTAL DISTRIBUTION OF TEMPERATURE

CHARACTERISTICS OF ISOTHERMS

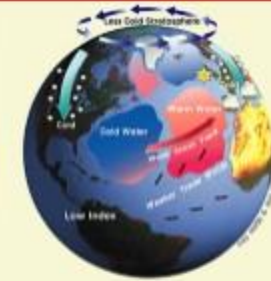


Picture credits: NCERT

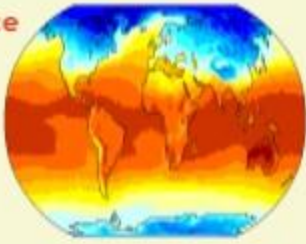
- 1 They run east-west and are generally parallel to latitudes.
- 2 This shows the control of latitudes on the horizontal distribution of temperature.
- 3 Generally they run straight but bend at the junction of continents and oceans.
- 4 Irregular in the northern hemisphere because of large extent of continents but they are more regular in the southern hemisphere due to over-dominance of oceans.



HORIZONTAL DISTRIBUTION OF TEMPERATURE

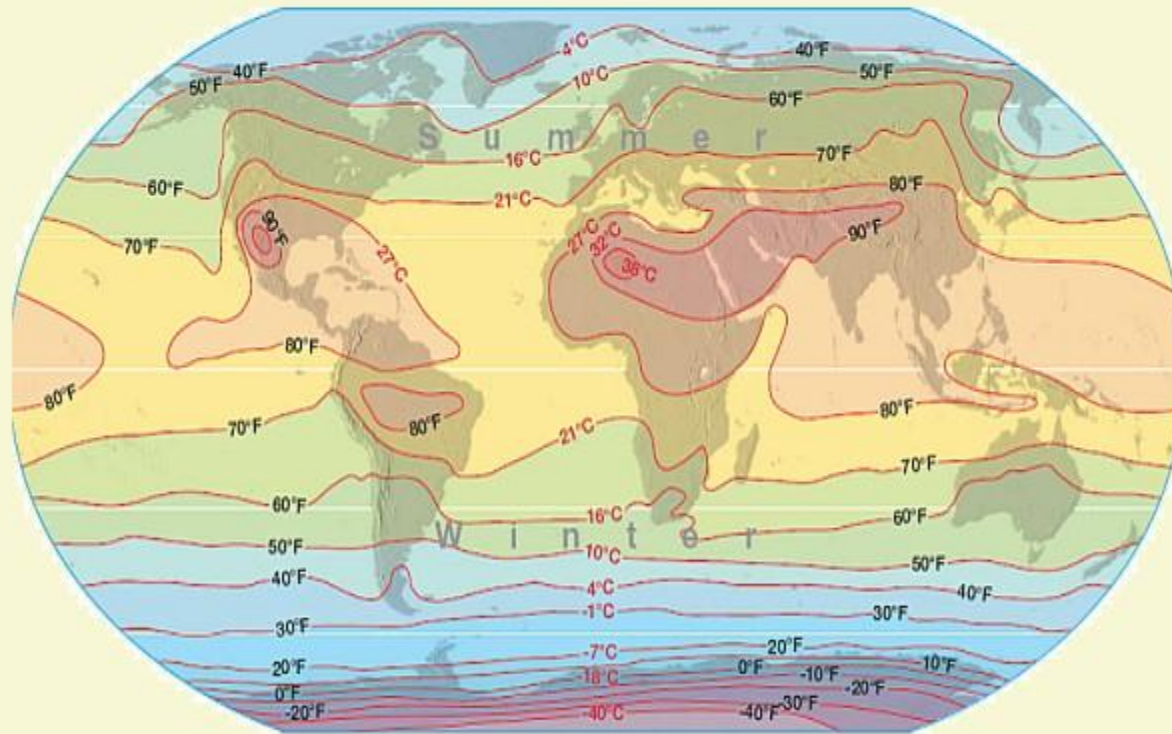
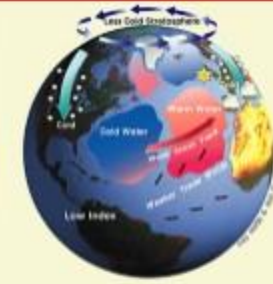


Southern Hemisphere dominated by oceans



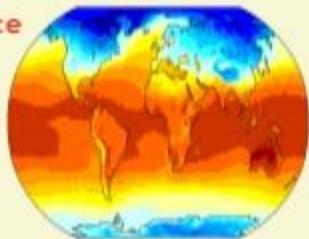
HORIZONTAL DISTRIBUTION OF TEMPERATURE

CHARACTERISTICS OF ISOTHERMS

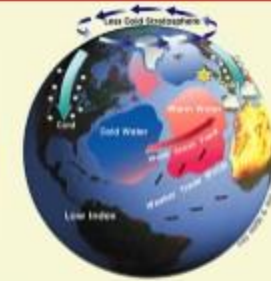


5 The **closely spaced isotherms** denote **rapid rate of change** of temperature and steep temperature gradient.

6 On the other hand, **widely spaced isotherms** indicate **slow rate** of temperature change and low temperature gradient.

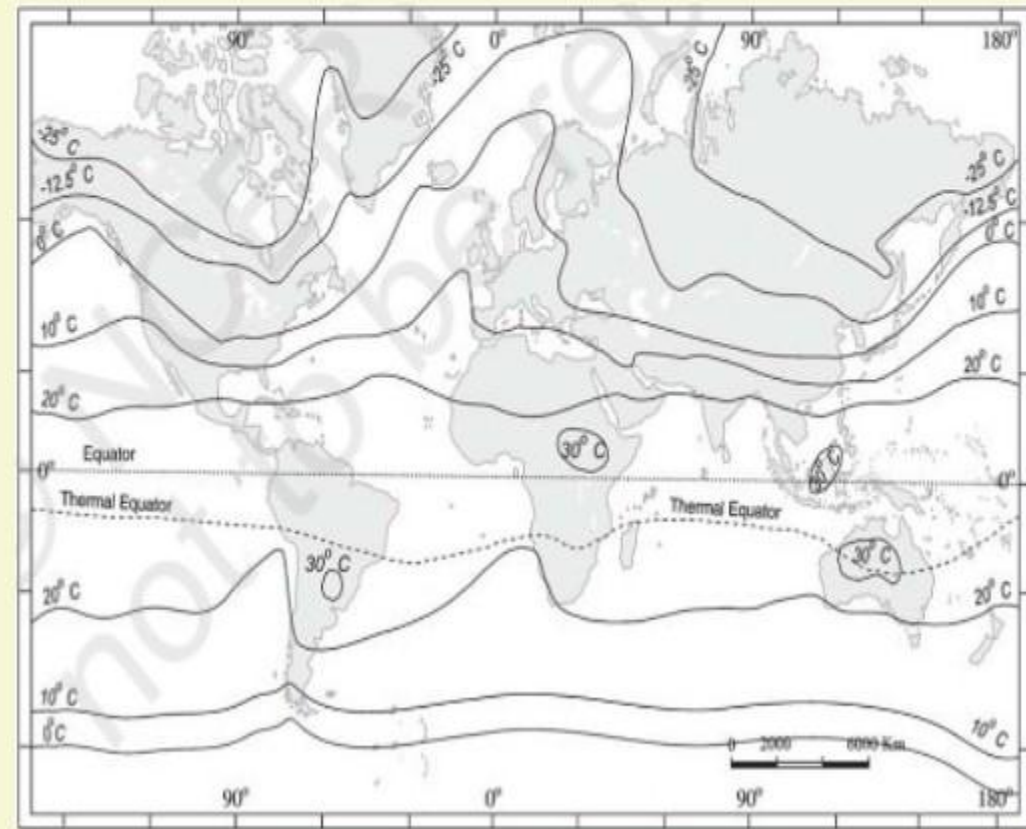


HORIZONTAL DISTRIBUTION OF TEMPERATURE



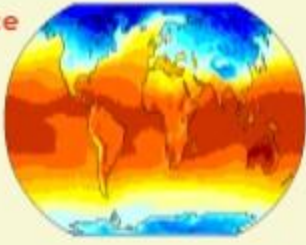
DISTRIBUTION OF ISOTHERMS IN JANUARY

WINTERS IN NORTHERN HEMISPHERE

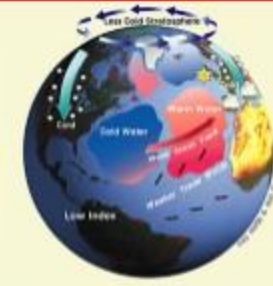


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- 1** Isotherms in winter in...
Continents = bend equatorwards
Oceans = bend polewards
- 2** The moderating effect of the ocean makes the isotherm to bend northwards over oceans.
- 3** The continentality effect in continents makes them much cooler than the corresponding ocean part of the same latitude.



HORIZONTAL DISTRIBUTION OF TEMPERATURE



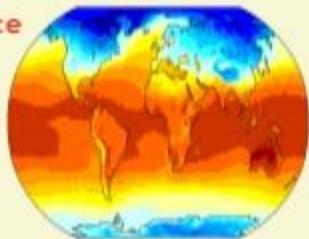
DISTRIBUTION OF ISOTHERMS IN JANUARY

WINTERS IN NORTHERN HEMISPHERE

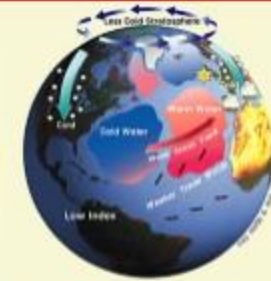


Isotherms bend
towards pole in oceans

Isotherms bend
towards equator in continents

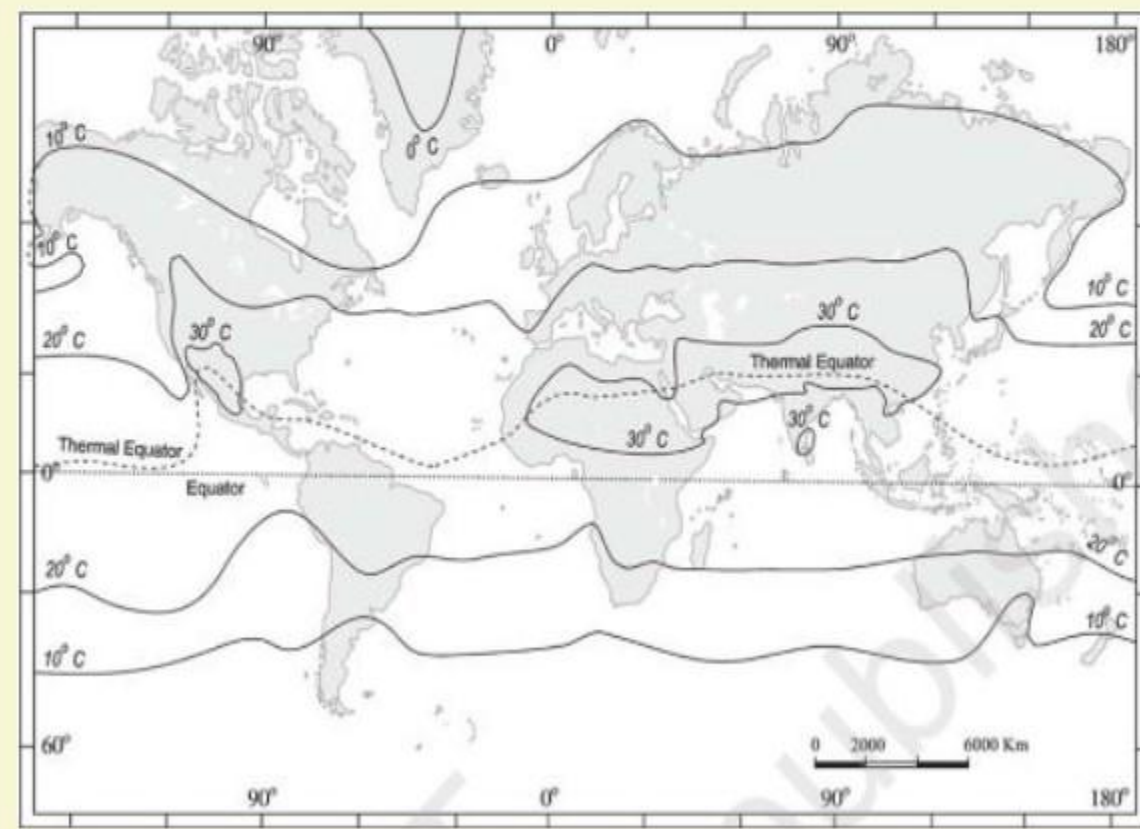


HORIZONTAL DISTRIBUTION OF TEMPERATURE



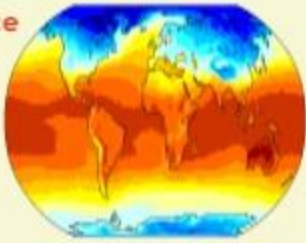
DISTRIBUTION OF ISOTHERMS IN JULY

SUMMERS IN NORTHERN HEMISPHERE



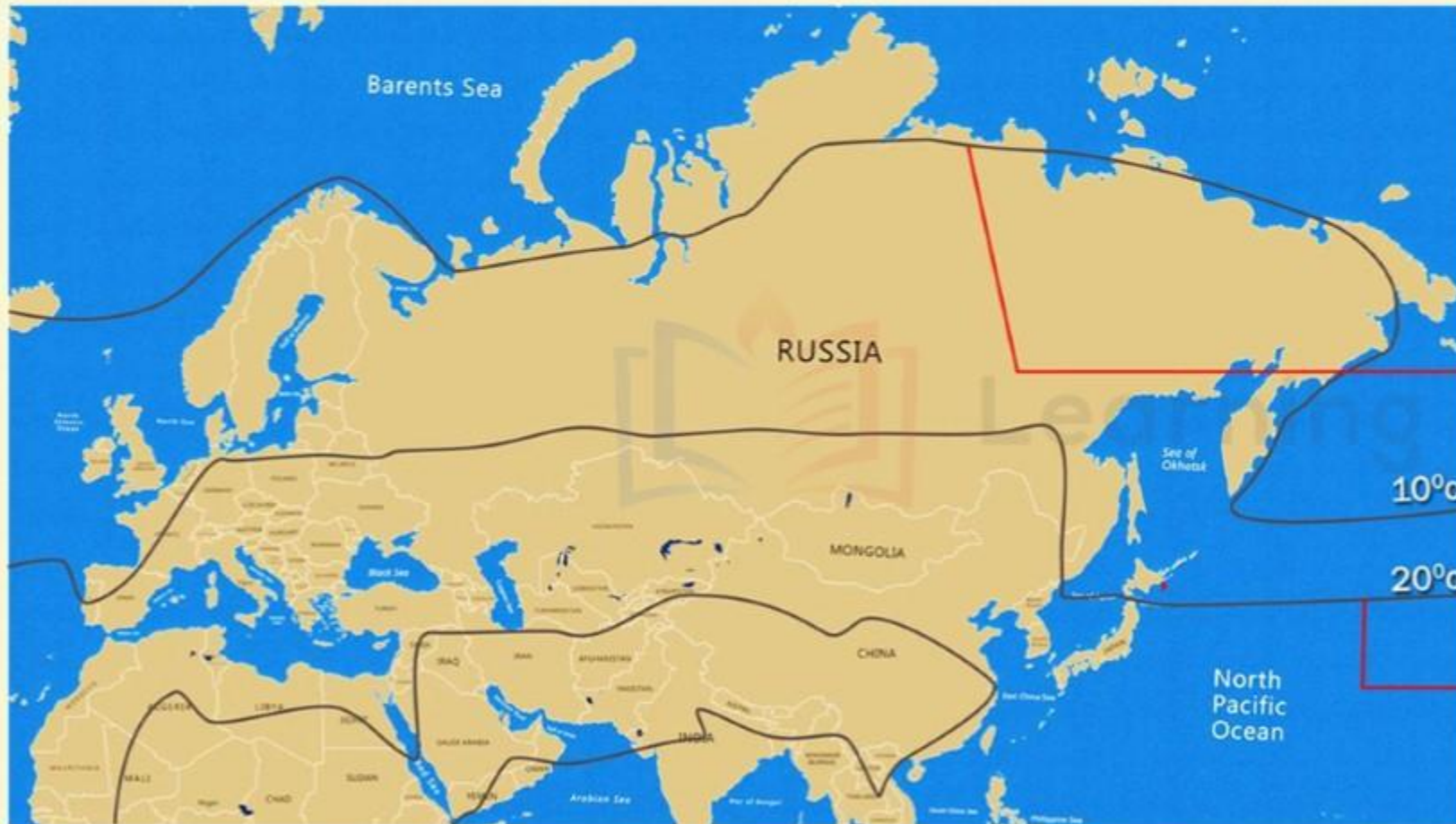
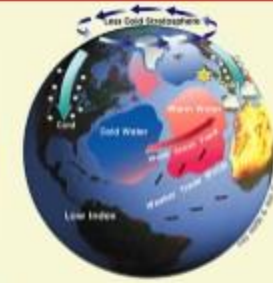
- 1 Isotherms in summer in...
Continents = bend polewards
Oceans = bend equatorwards
- 2 The moderating effect of the ocean makes the isotherm to bend southwards
- 3 The continentality effect in continents makes them much hotter than the corresponding ocean part of the same latitude.

Picture credits: NCERT



HORIZONTAL DISTRIBUTION OF TEMPERATURE

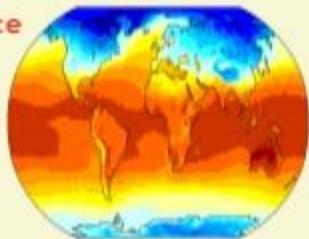
DISTRIBUTION OF ISOTHERMS IN JULY



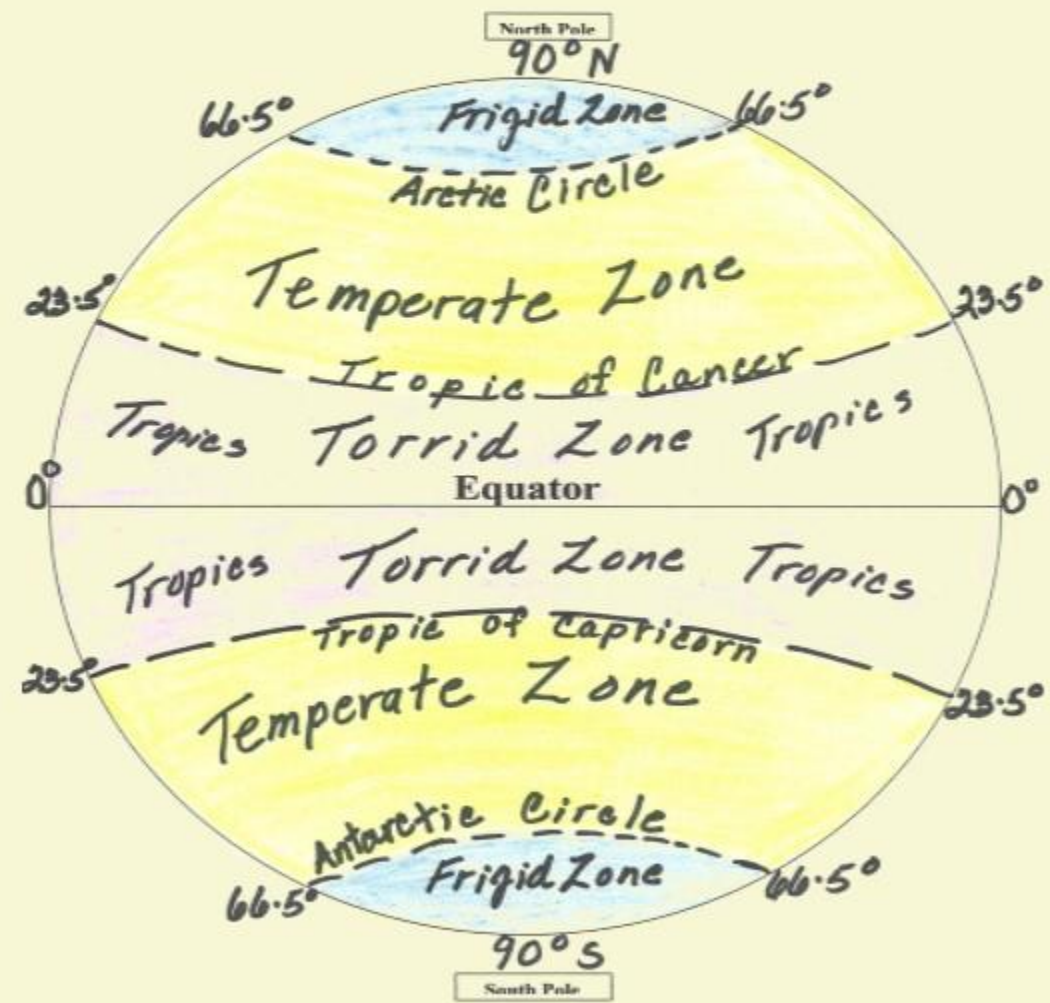
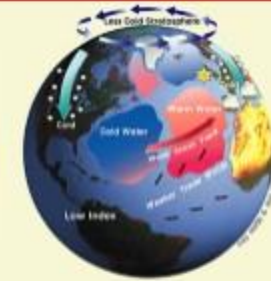
SUMMERS IN NORTHERN HEMISPHERE

Isotherms bend polewards in continents

whereas in oceans they bend equatorward

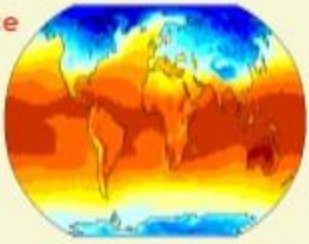


REGIONAL DISTRIBUTION OF TEMPERATURE

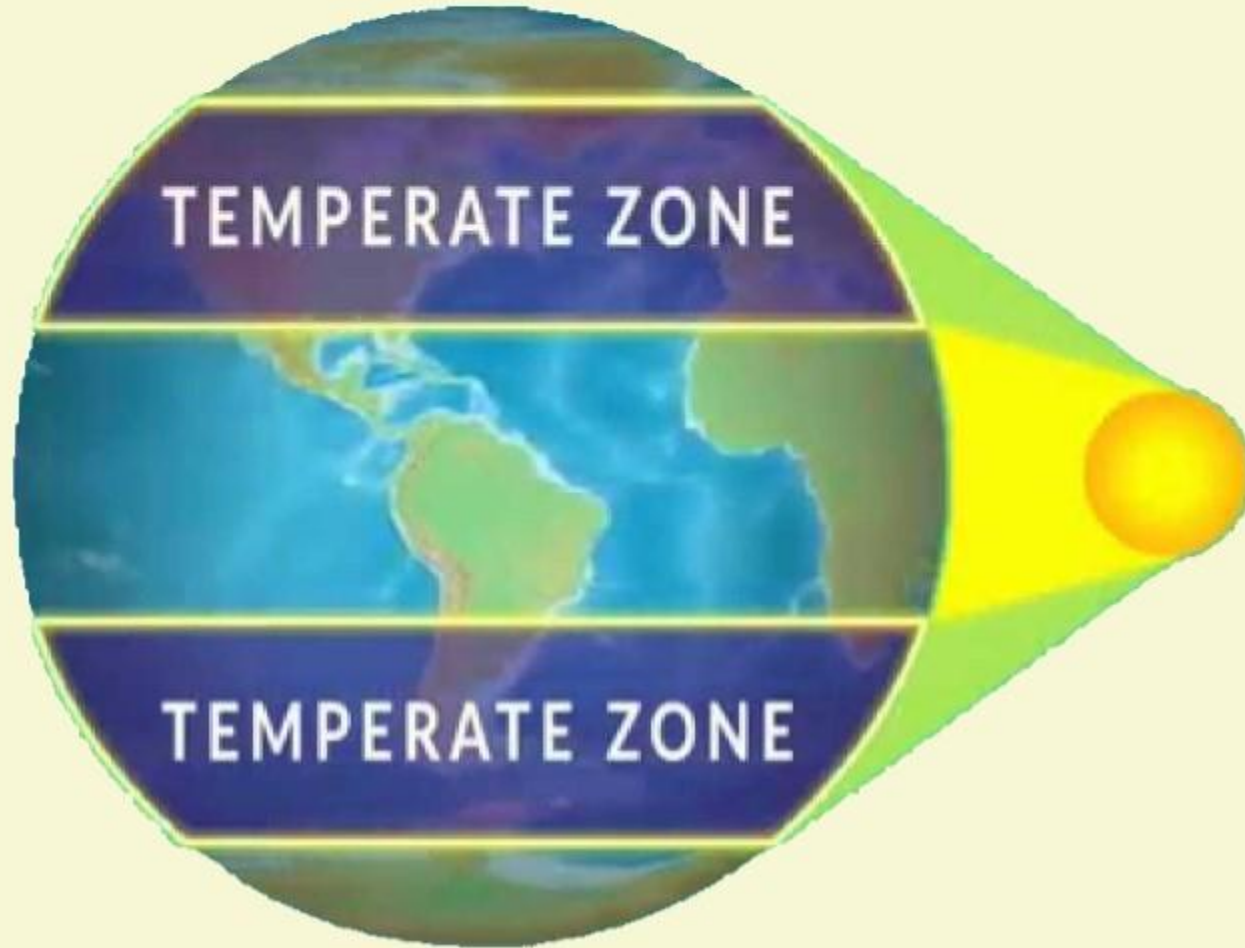
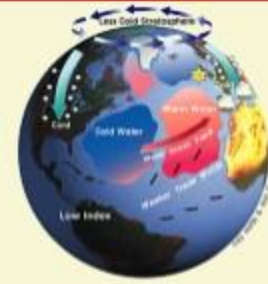


TROPICAL ZONE

- 1 Tropical zone extends between 23.5°N & 23.5°S .
- 2 Vertical rays of the Sun fall on equator throughout the year.
- 3 No winter around the equator because of high temperature prevailing throughout the year.
- 4 But as one approaches the tropics of Cancer and Capricorn, summer and winter are clearly observed and differentiated.

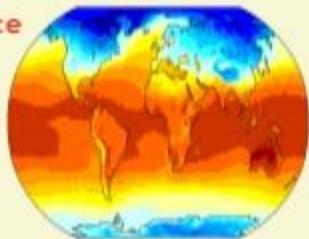


REGIONAL DISTRIBUTION OF TEMPERATURE

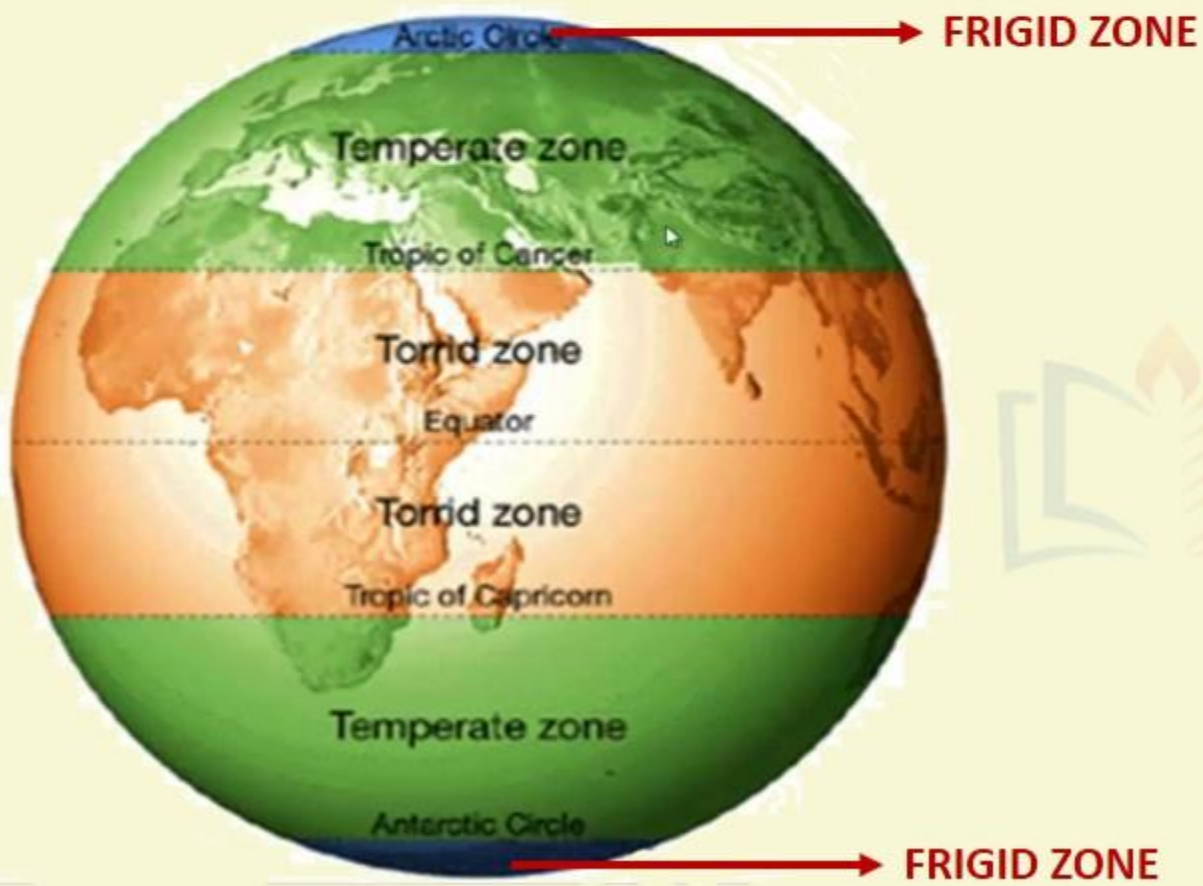
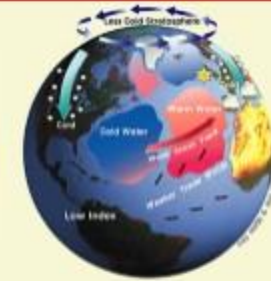


TEMPERATE ZONE

- 1** Temperate zone extends between 23.5° & 66.5° in both the hemispheres.
- 2** There are marked seasonal contrasts with the northward and southward march of the sun.
- 3** This migration of the overhead sun will cause a wide variation of temperature between summers and winters.

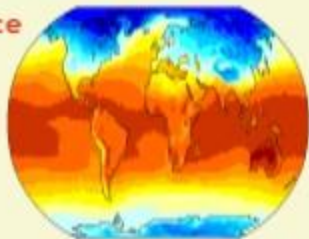


HORIZONTAL DISTRIBUTION OF TEMPERATURE

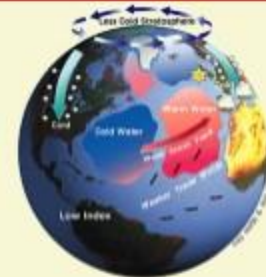


FRIGID ZONE

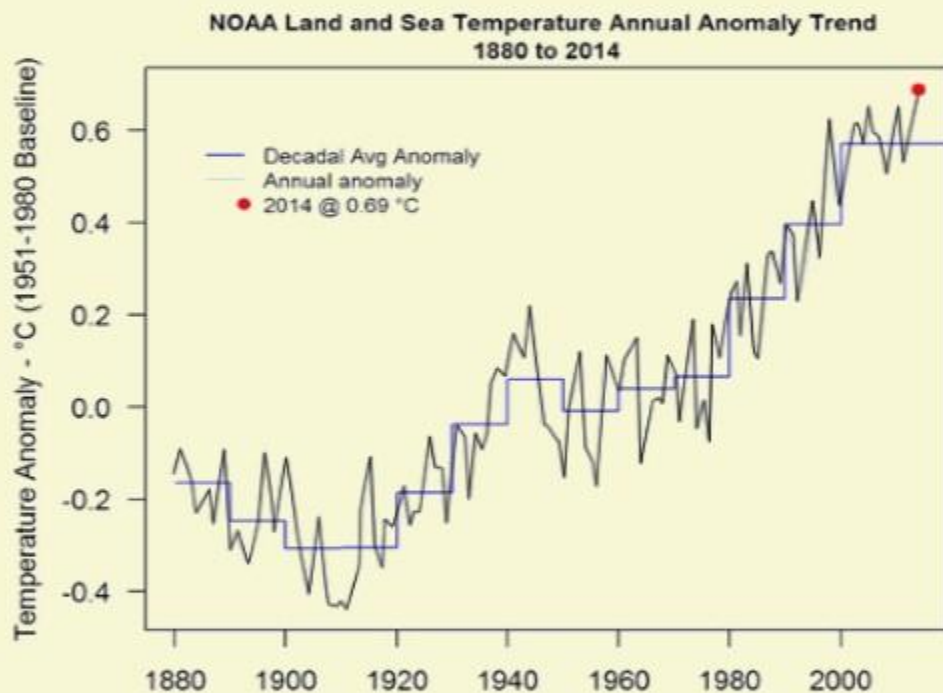
- 1 Frigid zone extends between 66.5° & poles in both the hemispheres.
- 2 It is characterised by more oblique rays of the sun throughout the year resulting in exceptionally low temperatures.
- 3 Sun is never vertical and the ground is covered with snow as temperature remains more or less below freezing point.



HORIZONTAL DISTRIBUTION OF TEMPERATURE



THERMAL ANAMOLY



1 The difference of observed temperature of a place and the mean temperature of the latitude passing through that place is called **thermal anomaly**.

2 If the observed temperature of a particular place is more than the mean temperature of the latitude of that place, the thermal anomaly is called **positive thermal anomaly**.

3 If the observed temperature of a particular place is less than the mean temperature of the latitude of that place, the thermal anomaly is called **negative thermal anomaly**.

4 The equal thermal anomaly of several places is called **isanomalous temperature** and the lines drawn on the world map joining places of **equal thermal anomaly** are called **isoanamols**.

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